

Task 3 Importing and creating Python modules and packages in Python

Date: 20/8/21

Program:

Aim: To implement and demonstrate the process of importing built-in modules, creating user-defined modules, and organizing code into packages in Python, thereby promoting code reusability, modularity and maintainability.

Algorithm:

1. Import required modules: math, random, os, sys, statistics, pathlib

2. math & random:

- compute $\sqrt{5}$, radians(30), a random float in [0, 0.1], a random integer in [2, 6] inclusive, π , ceil(2.3), floor(2.3), factorial(5), gcd(5, 15), abs(-10), pow(3, 5), log base 3 of 2, $\log_{10}(a)$ for $a=100$, and check NaN/infinity

3. OS & sys:

- create C:\pythonlab if not present and print the current working directory.
- create c:\pythonlab\214 if not present and change the current working directory to it
- list all files/directories in the new current directory
- print Python interpreter version.

4. statistics:

- on lists: [5, 6, 8, 10] and [2, 5, 3, 2, 8, 3, 9, 4, 2, 5, 6] compute mean, median, mode, stdev.

5. Print neatly for formatted results.


```

import math
import random
import os
import sys
import statistics as stats
from pathlib import Path

Print("\n --- MATH & RANDOM ---")

Print("sqrt(5) = ", math.sqrt(5))
Print("radians (30) = ", math.radians(30))
Print("random() in [0,1) = ", random.random())
Print("randint(2,6) = ", random.randint(2,6)) # inclusive
Print("pi = ", math.pi)
Print("ceil(2.3) = ", math.ceil(2.3))
Print("floor(2.3) = ", math.floor(2.3))
Print("factorial(5) = ", math.factorial(5))
Print("gcd(5,15) = ", math.gcd(5,15))
Print("abs(-10) = ", abs(-10))
Print("pow(3,5) = ", pow(3,5))
Print("log base 3 to 2 = ", math.log(2,5))
a-val=100
Print("log 10 ({a-val}) = ", math.log(a-val))
inf-val=float('inf')
nan-val=float('nan')
Print("isinf(oo) = {math.isinf(inf-val)}, isnan(NaN) = {math.isnan(nan-val)}")

Print("\n --- OS & sys ---")
path-pythonlab=Path(r"C:\Python lab")
path-pythonlab.mkdir(parents=True, exist_ok=True)
Print("created/successful: {path-pythonlab}")
Print("current working directory: ", os.getcwd())
target-dir=Path(r"C:\Python slot 5214")
target-dir.mkdir(parents=True, exist_ok=True)
os.chdir(target-dir)

Print("changed into: {target-dir}")
Print("Directory content: ", os.listdir())
Print("Python version: ", sys.version)

```


Expected sample output

-- MATH & RANDOM --

$\text{sqrt}(5) = 2.2360679774$

$\text{radians}(36) = 0.6283185307$

$\text{random}() \text{ in } [0,1) = 0.3744487175$

$\text{randint}(2,6) = 6$

$\pi = 3.14159265358$

$\text{ceil}(2.3) = 3$

$\text{floor}(2.3) = 2$

$\text{factorial}(5) = 120$

$\text{gcd}(5, 15) = 5$

$\text{abs}(-10) = 10$

$\text{pow}(3, 5) = 243$

$\log \text{ base } 3 \text{ of } 2 = 0.6309297525$

$\log_{10}(100) = 2.0$

$\text{is_int}(\infty) = \text{True}$, $\text{is_nan}(\text{NaN}) = \text{True}$.

-- OS & SYS --

Created / ensured = $C:\backslash \text{python lab}$

Current working directory: $C:\backslash \dots$ (your current path)

Created / ensured & changed into = $C:\backslash \text{python lab} \text{ s214}$

Directory contents of $C:\backslash \text{python lab} \text{ s214}$: $[\]$

Python version: 3.x.x (--- details ---)

-- STATISTICS --

$\text{mean}([5, 6, 8, 10]) = 7.25$

$\text{median}([5, 6, 8, 10]) = 7.0$

$\text{mode}([2, 5, 3, 2, 8, 3, 9, 2, 5, 6]) = 2$

$\text{stdev}([2, 5, 3, 2, 8, 3, 9, 4, 2, 5, 6]) = 2.27156338201093$


```

Print("\n --- STATISTICS ---")
data 1 = [5, 6, 8, 10]
data 2 = [2, 5, 3, 2, 8, 3, 9, 4, 2, 5, 6]
Print (f"mean ({data1}) = ", stats.mean(data1))
Print (f"median ({data 2}) = ", stats.median(data1))
Print (f"mode ({data 2}) = ", stats.mode(data 2))
Print (f"stdev ({data 2}) = ", stats.stdev(data 2))

```

--- MATH & RANDOM ---

```

sqrt(5) = 2.236 067 977 4 979
radians(30) = 0.523 598 775 5 982 988 < will vary
radians(2, 6) = 6
pi = 3.141 592 653 5 897 93
Ceil(2.3) = 3
Floor(2.3) = 2
factorial(5) = 120
gcd(5, 15) = 5
abs(-10) = 10
pow(8, 5) = 32768
log base 3 of 2 = 0.630 929 753 571 457 4
log10(100) = 2.0
isinf(oo) = True, isnan(NaN) = True

```

--- OS & sys ---

```

created/ensured: C:\python lab
current working directory: C:\... (your current path)
created/ensured & changed into: C:\Python slot Set4
Directory contents of C:\python slot Set4
python version: 3.x.x (--- details ---)

```

--- STATISTICS ---

```

mean([5, 6, 8, 10]) = 7.25
median([5, 6, 8, 10]) = 7.0
mode([2, 5, 3, 2, 8, 3, 9, 4, 2, 5, 6]) = 2
stdev([2, 5, 3, 2, 8, 3, 9, 4, 2, 5, 6]) = 2.271 563 338 3201

```

Output

Program:

cardfun

import random

def func():

cards = []

for i in range(1, 53):

cards.append(i)

shuffled = cards = random.sample(cards, len(cards))

print("\n\n", shuffled - cards, "\n\n")

myMod.py

import cardfun

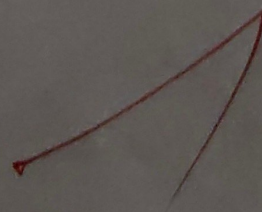
cardfun.func()

output:

RESTART:

C:\Users\student.MATE\CG833\AppData\Local\Programs\Python\Python38-64\Lib\site-packages\card\pack\myMod.py.

[5, 24, 13, 22, 20, 41, 38, 51, 4, 7, 34, 29, 14, 60, 37, 40, 15, 35, 17, 18, 33, 39, 36, 44, 12, 6, 16, 19, 48, 29, 2, 27, 11, 31, 46, 28, 91, 32, 8, 25, 30, 23, 26, 10, 43, 47, 3, 44, 52, 1, 45, 9].



Date: 20/8/25

3.3.

You are tasked with developing a modular calculator application in Python. The calculator should support basic arithmetic operations: addition, subtraction, multiplication and division. Each operation should be implemented in a separable module. Additionally, you should create a main program to handle user input, call the appropriate module, and display the results.

Algorithm:-

1. Define functions for addition, subtraction, multiplication and division
2. Handle division by zero by raising an error if the divisor is zero.
3. Import the module (my math) containing these functions.
4. Initialize two numbers ($a=10$, $b=5$)
5. Call each function using `my math <function-name> (a,b)`
6. Print the results of all operations.

Program(my)math

```
def add(a,b):
```

```
    return a+b
```

```
def subtract(a,b):
```

```
    return a-b
```

```
def multiply(a,b):
```

```
    return a*b
```

```
def divide(a,b):
```

```
    if b==0:
```

```
        raise ValueError("cannot divide by zero")
```

```
    return a/b
```

```
import mymath
```

```
a=10
```

```
b=5
```

```
print("Addition :", mymath.add(a,b))
```

```
print("subtraction:", mymath.subtract(a,b))
```

```
print("multiplication:", mymath.multiply(a,b))
```

```
print("Division :", mymath.divide(a,b))
```

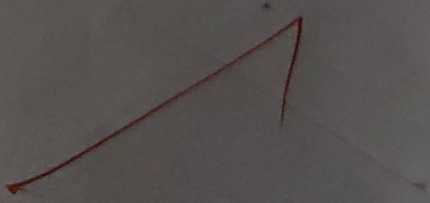
Output

Addition: 15

subtraction: 5

multiplication: 50

Division: 2.0



3.4

You are working on a python project that requires you to perform various mathematical operations and geometric area calculations. To organize your code better, you decide to create a package named my package which includes sub packages pack1 and pack2 with two modules: math functions and area functions. Demonstrate the use of the function by performing a few calculations and printing the results.

Algorithm:

1. create mathfunctions.py module:
2. create area functions.py module:
3. create main.py:
4. print the output as expected.

Program:

1. Create the mathfunctions.py module:

```
def add(a,b):
    return a+b

def subtract(a,b):
    return a-b

def multiply(a,b):
    return a*b

def divide(a,b):
    if b == 0:
        return "Error! Division by zero".
    return a/b
```

2. create the area functions.py module.

```
import math

def circle-area(radius):
    return math.pi * radius * radius

def rectangle-area(length,width):
    return length * width

def triangle-area(base,height):
```


Output

-----RESTART 'C':

Addition: 15

subtraction: 5

multiplication: 50

Division: 2.0

Circle Area (radius=7): 153.93804002589985

Rectangle Area (5x6): 50

Triangle Area (base=6, height=8): 24.0

return 0.5 * base * height

3. Create the main.py file

import math function

import area function.

using math functions

Print ("Addition:", math function, add(10,5))

Print ("Subtraction:", math function, subtract(10,5)).

Print ("multiplication:" math function, multiply(10,5))

Print ("Division:", math function, divide(10,5))

using area functions.

Print ("Circle Area (radius = 7):", area function, Circle-area(7))

Print ("Rectangle Area(5x10):", area function, rectangle-area(5,10))

Print ("Triangle Area (base = 6, height = 8):", area function, triangle-area(6,8))

VEL TECH	
EX NO.	3
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	
DATE WITH DATE	15

Result: Thus, the program importing python modules and packages was successfully executed and the output was verified.